3(a)(i)

The heat required to melt the 5.0 kg block of ice, Q:
= mass × latent heat of fusion
= 5.0 kg × 3.34 x 10^5 J kg^{-1}
= 1.67 x 10^6 J

C1

Time to melt the ice:
= Q / P
= 1.67 x 10^6 / 4180
= 400 s

A1

3(a)(i)

Volume of the ice, V_{ice}
= mass of ice / density of ice
= 5.0 / 0.915
= 5.4645 m^3

Volume of the liquid water, V_{water}
= mass of water / density of water
= 5.0 / 0.999
= 5.0050 m^3

Change in volume, ΔV
= 5.01 m^3 - 5.46 m^3
= -0.4595 m^3

C1

Work done ON ice, w
= - P x ΔV
= - [1.01 x 10^5 * (-0.4595)]
= 4.64 x 10^4 J or 4.6 x 10^4 J

C1
A1

3(b)

As the ice melts, its internal energy increases due to the energy used to increase the intermolecular separation. The source of this energy is the heat supplied to the ice and the work done on the ice by the atmospheric pressure (since the ice contracts as it melts)

As the liquid water increases in temperature, its internal energy increases. The source of this energy is the heat supplied to the water from the surroundings and the further contraction of the volume of the water (as its density continues to increase) means that the atmospheric pressure does further work on the water.

Process	*ΔU*	*q*	*w*
Ice melting into liquid water at 0°C	**positive**	**positive**	**positive**
Liquid water warming up from 0°C to 4°C	**positive**	**positive**	**positive**

A1 (column *ΔU* and *q*)

A1 (column w)